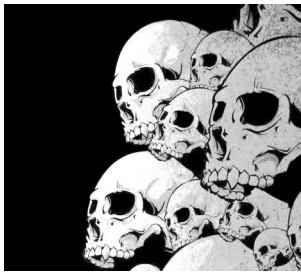


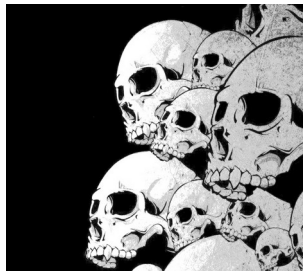
Y. Collette (ycollette.nospam@free.fr)
<https://audinux.github.io/>



Licenses Code

	Utiliser	Redistribuer	Modifier
Propriétaire			
Shareware			
Freeware			
Logiciel Libre			

	Code source disponible	Utilisation libre du programme	Copie du programme	Distribution du programme	Modification du programme	Paiement de royalties	Obligation de rendre libre toute modification	Compatibilité avec la GPL
BSD	oui	oui	oui	oui	oui	gratuit	non	non
Apache	oui	oui	oui	oui	oui	gratuit	oui	non
Mozilla	oui	oui	oui	oui	oui	gratuit	non	non
Netscape	oui	oui	oui	oui	oui	gratuit	non	non
Sun	oui	oui	oui	oui	oui	gratuit	oui	non
IBM	oui	oui	oui	oui	oui	gratuit	oui	non
PHP	oui	oui	oui	oui	oui	gratuit	oui	non



Licenses Creative Commons

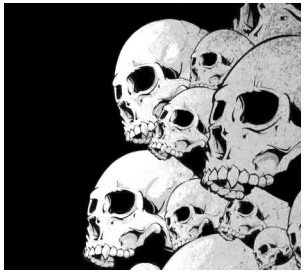
A license for
documents

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Licence CC	Bouton	Explications	Partage permis ?	Exploitation permise?	Remix permis?
Attribution	BY	- On doit citer QUI est l'auteur de l'œuvre originale. - L'utilisation commerciale de l'œuvre est permise. - Vous pouvez remixer l'œuvre. - Le partage de l'œuvre est (toujours) permis.			
Share Alike Partage à l'identique	BY SA	- On doit citer QUI est l'auteur de l'œuvre originale. - L'utilisation commerciale de l'œuvre est permise. - Vous pouvez remixer l'œuvre. - Cette licence doit toujours être utilisée sur toutes vos versions dérivées de l'œuvre originale. - Le partage de l'œuvre est (toujours) permis.			
No Derivative Modification non permise	BY ND	- On doit citer QUI est l'auteur de l'œuvre originale. - L'utilisation commerciale de l'œuvre est permise. - Vous NE pouvez PAS remixer l'œuvre. - Le partage de l'œuvre est (toujours) permis.			
Non-Commercial Usage commercial non permis	BY NC	- On doit citer QUI est l'auteur de l'œuvre originale. - L'utilisation commerciale de l'œuvre n'est PAS permise. - Vous pouvez remixer l'œuvre. - Le partage de l'œuvre est (toujours) permis.			
Non-Commercial Usage commercial non permis + Share Alike Partage à l'identique	BY NC SA	- On doit citer QUI est l'auteur de l'œuvre originale. - L'utilisation commerciale de l'œuvre n'est PAS permise. - Vous pouvez remixer l'œuvre. - Cette licence doit toujours être utilisée sur toutes vos versions dérivées de l'œuvre originale. - Le partage de l'œuvre est (toujours) permis.			
Non-Commercial Usage commercial non permis + No Derivative Modification non permise	BY NC ND	- On doit citer QUI est l'auteur de l'œuvre originale. - L'utilisation commerciale de l'œuvre n'est PAS permise. - Vous NE pouvez PAS remixer l'œuvre. - Le partage de l'œuvre est (toujours) permis.			





Free software



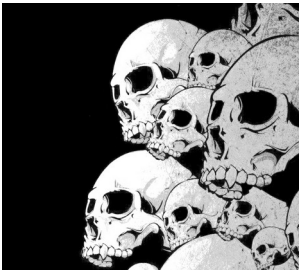
The good



The brute

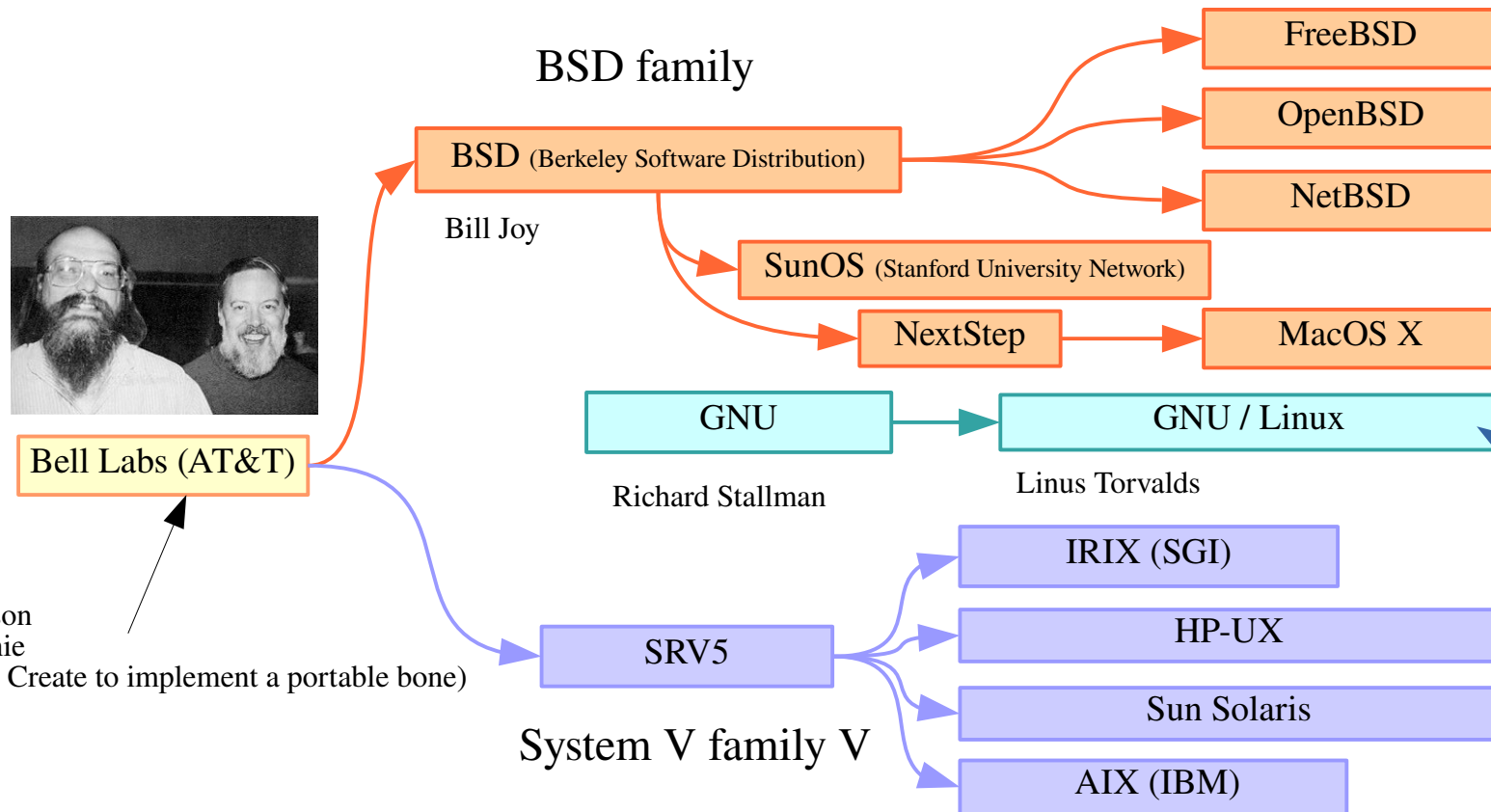


The Crook



UNIX family tree

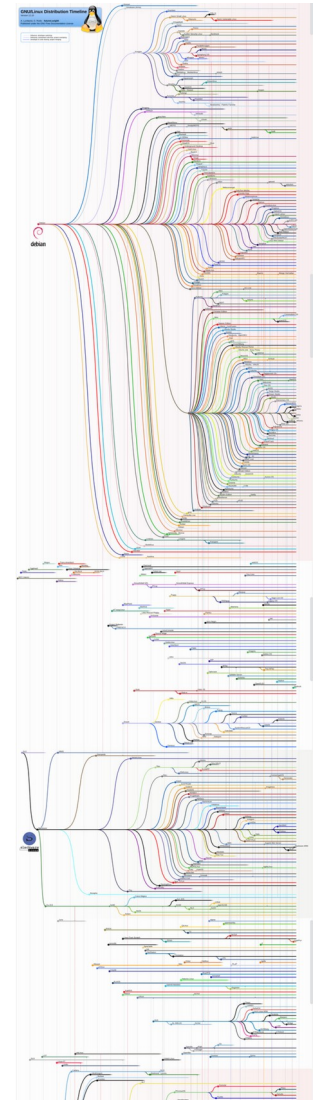
1970 1980 1990 2000 Time →

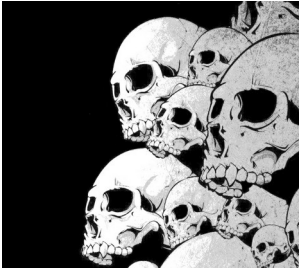


Distributions by tens

And ...
Kxstudio
Dreamstudio
APODIO GNU
Linux
Librazik

Lots of
distributions for
computer
-assisted music





How to test Linux ?

Either we install Linux directly on the PC (to replace Windows or Dual Boot)

Either we use a virtual machine:

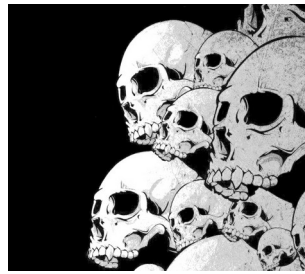
It's a PC in the PC

Almost the same performance as a standard PC

But not suitable for the Mao (too much latency)

But that will be enough to test ...



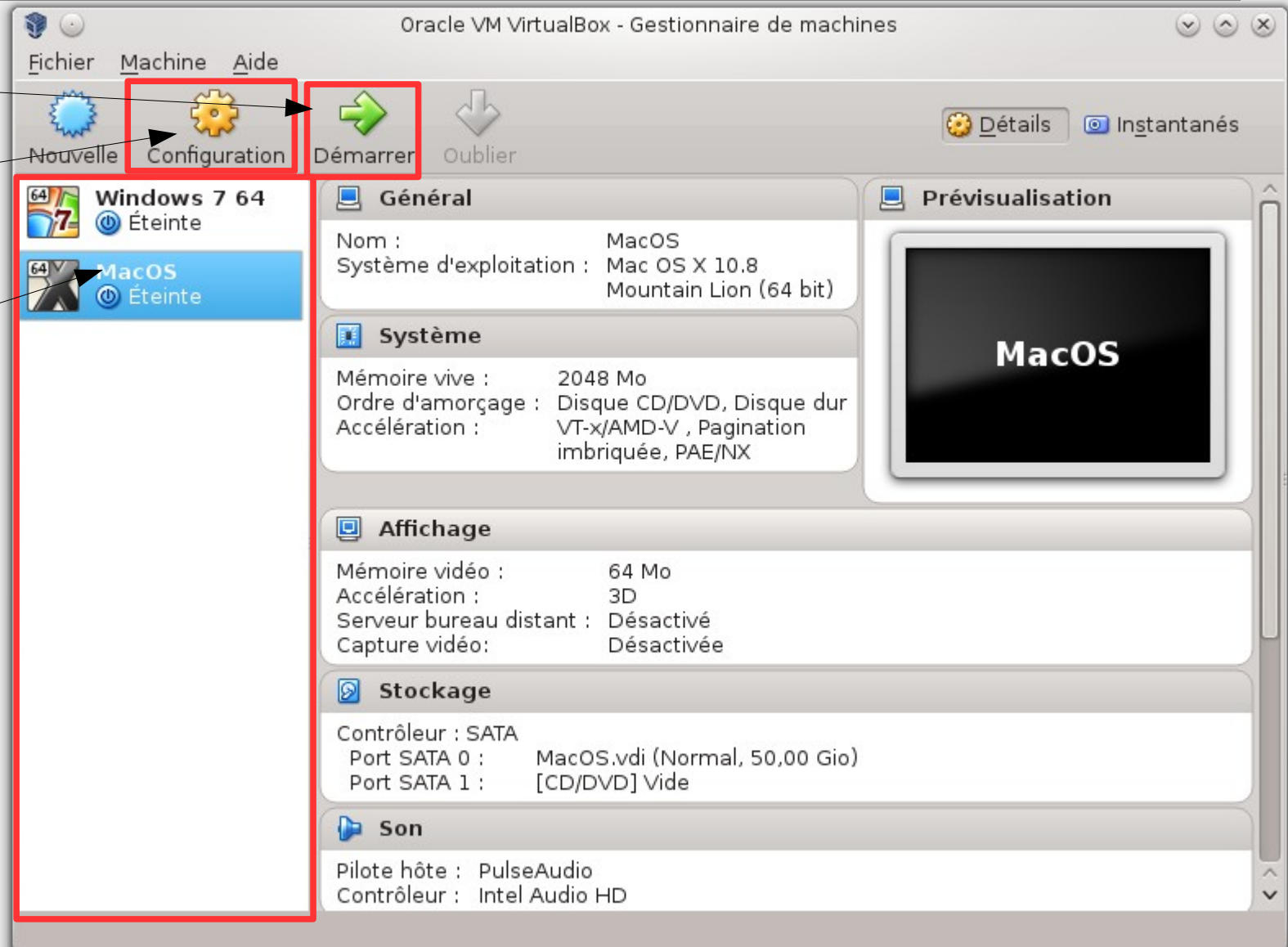


VirtualBox

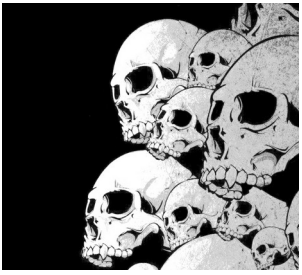
Startup

Configuration

List of virtual machines



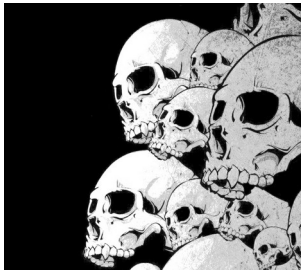
24/08/2013



Update

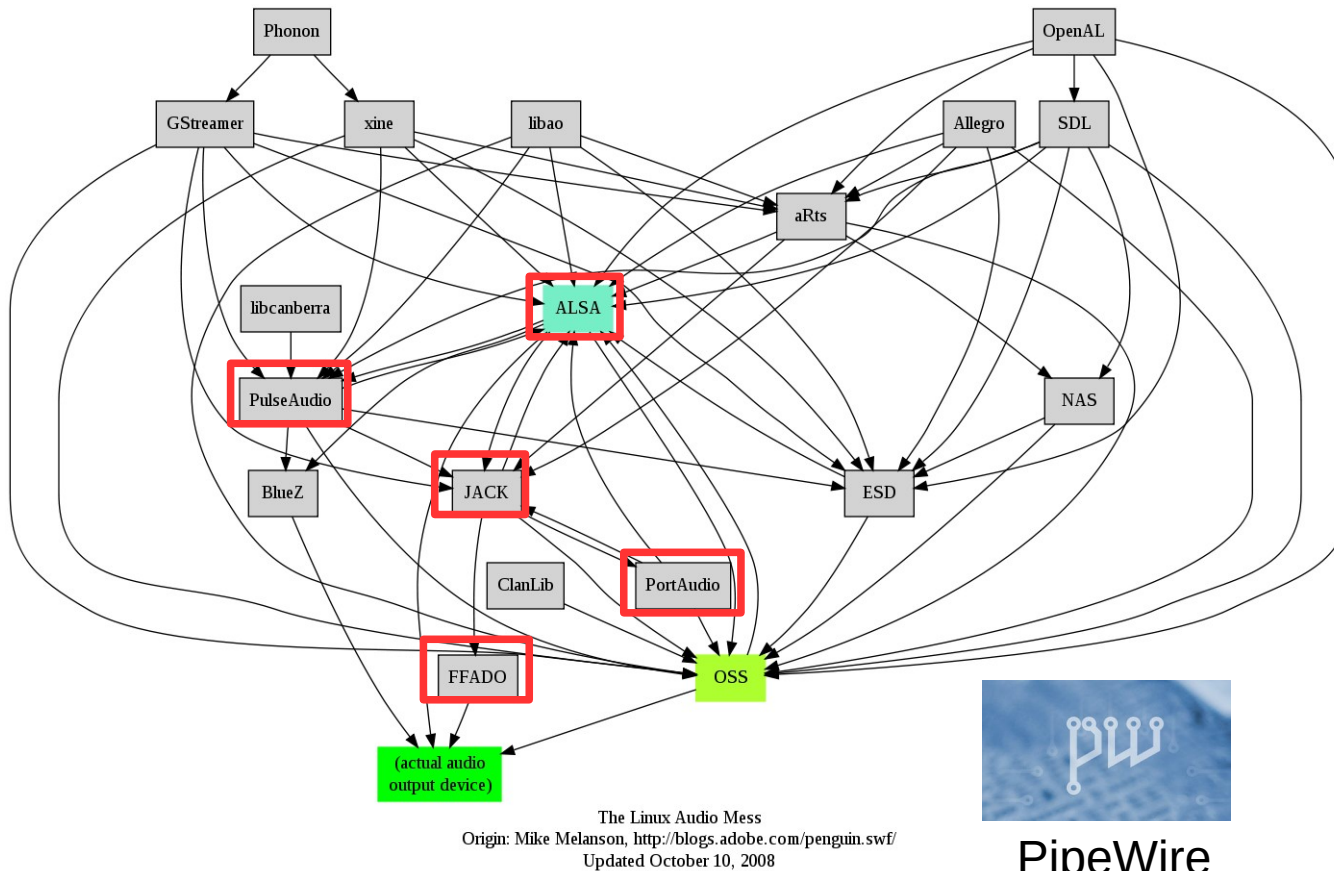
When the Kernel Linux is updated, it is necessary to update the Linux Additions Virtualbox. Under the host Linux, you have to mount the CDROM, then go to the /Run/Media /.../ Vbox

```
$ su  
# Go to the /Run/Media/..... directory /..  
$ ./Vboxlinuxadditions
```

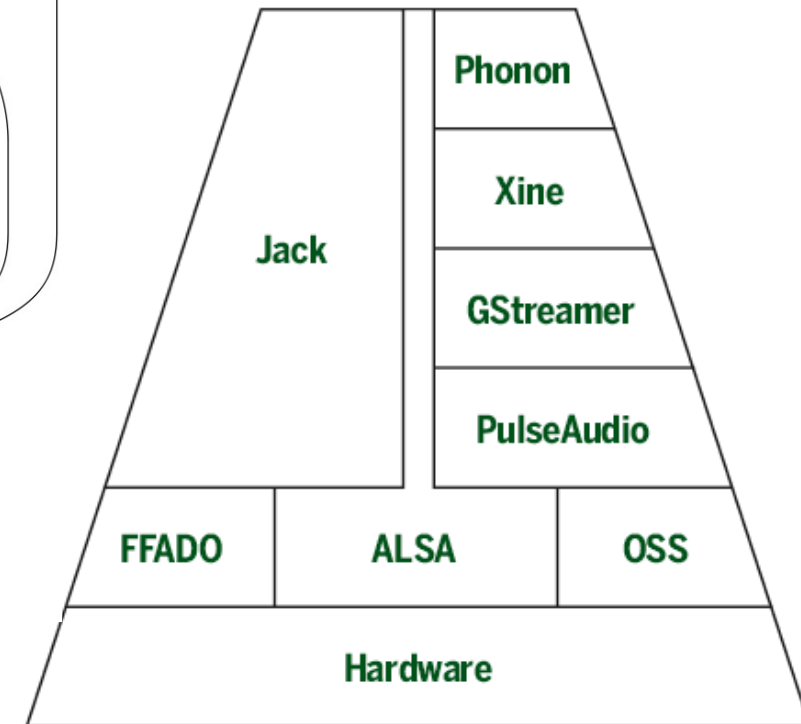


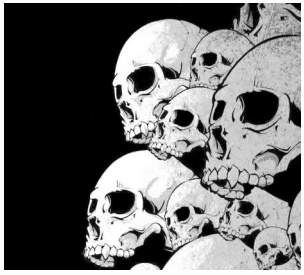
Linux audio organization

The Linux audio problem



PipeWire





Linux Audio: equipment

Before buying equipment, have the Google reflex.

There is also <http://linuxmao.org>

Check that your equipment is compatible before making your purchase.

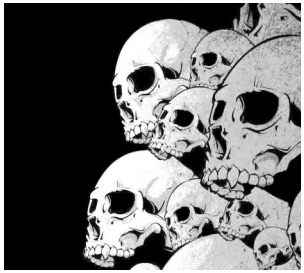
Especially if you buy an audio interface
Firewire: <https://www.ffado.org>

For USB sound cards, the best:

USB Class Compliant

These cards have a generic support under Linux





Linux Audio: equipment

Why buying a good preamp audio card?

Wikipedia response:

When you follow a first amplifier by a second, the latter amplifies the signal and the noise that the first sends it and he adds the noise he makes himself.

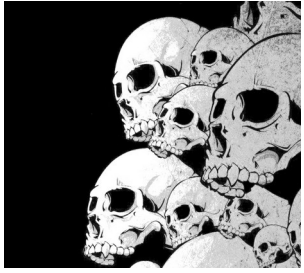
If the amplification chain includes n elements, the noise generated by the first floor is amplified by the stages 2, 3 ... n

Friis formula:
$$F = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 \cdot G_2} + \frac{F_4 - 1}{G_1 \cdot G_2 \cdot G_3} + \dots + \frac{F_n - 1}{G_1 \cdot G_2 \cdot G_3 \dots G_{n-1}}$$

where F_i is the noise factor (linear and not in db) of the i th element and G_i the gain (linear and not in db) of the i th element.

The formula tells us that it is the noise factor of the 1st element of the chain which largely determines the total noise factor. It is in particular for this reason that we use a low noise amplifier in the first floor of the reception chains of weak signals. This makes it possible to recover a sufficient level signal without degrading the signal ratio over noise too much, generally low at the input of the chain.

Example of calculation of the signal / noise ratio:
<http://f5zv.pagesperso-orange.fr/radio/rm/rm04/rm04c04.html>

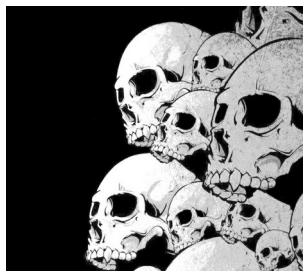


Initial settings

Configuration of the COPR Audinux repository

Definition of priorities, from memberships to groups

Installation of a real time nucleus (optional)



Fedora Repository configuration

Fedora: General distribution

CCRMA repository: A MAO application source for Fedora

CCRMA = Center for Computer Research in Music and Acoustics - Stanford

Installation of these repositories:

```
dnf install --nogpgcheck  
http://mirrors.ircam.fr/pub/planetccrma/mirror/fedora/linux/planetccrma/21/x86_64/  
planetccrma-repo-1.1-3.fc21.ccrma.noarch.rpm
```

```
dnf install --nogpgcheck  
http://mirrors.ircam.fr/pub/planetccrma/mirror/fedora/linux/planetccrma/21/x86_64/  
planetccrma-repo-testing-1.1-3.fc21.ccrma.noarch.rpm
```

RPM Fusion Free component:

```
dnf install --nogpgcheck http://fr2.rpmfind.net/linux/rpmsfusion/free/fedora/rpmsfusion-free-  
release-stable.noarch.rpm
```

RPM Fusion Non -free component:

```
dnf install --nogpgcheck http://fr2.rpmfind.net/linux/rpmsfusion/nonfree/fedora/rpmsfusion-  
nonfree-release-stable.noarch.rpm
```



Fedora Repository configuration

Audio tools in development version or not present in other repositories:

Version Fedora 42, 43, 44, rawhide:

```
$ dnf copr enable ycollet/audinux
```

Address of the COPR Audinux repository:

<https://copr.fedorainfracloud.org/coprs/ycollet/audinux>

Where to postpone the bugs linked to the Audinux COPR repository:

<https://github.com/audinux/fedora-spec>

Web site for Audinux :

<https://audinux.github.io>



Fedora

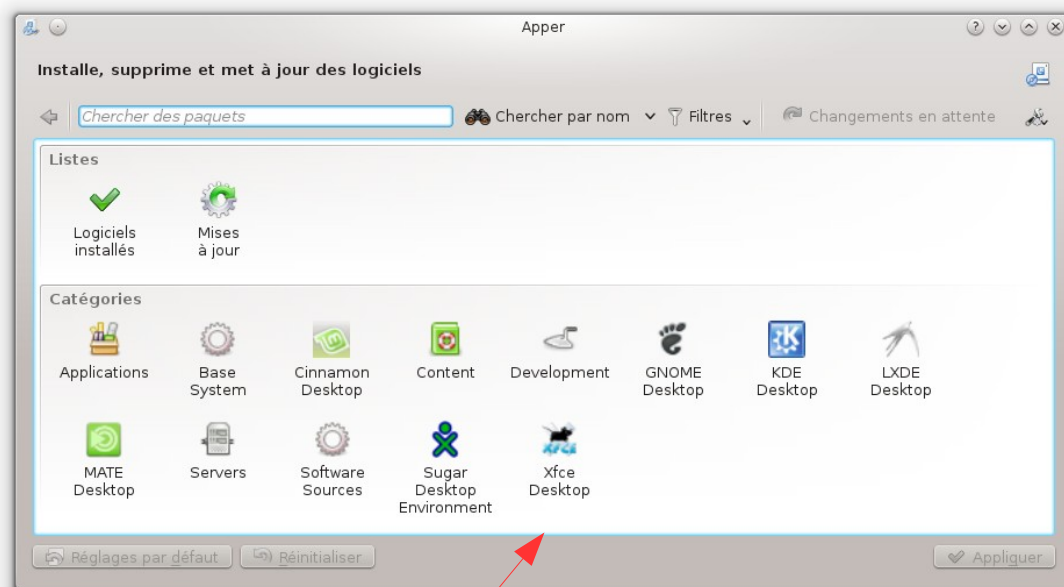
Package installation:

- Start menu → software management
- or via the 'DNF' command:

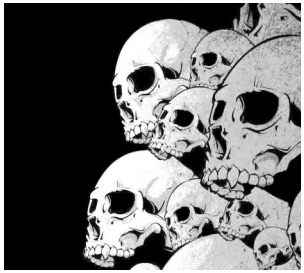
```
$ dnf search <application>  
$ dnf install <application>  
$ dnf remove <application>  
$ dnf list installed | grep <application>
```

Update :

```
$ dnf --refresh update
```



You can use Discover as a graphical interface



Fedora

Some settings

Install the real time kernel (Kernel-RT-optional).

The real -time kernel reduces latency, but is not mandatory. The standard kernel can have a low enough latency for the MAO.

Disable energy management:

```
# Become a "super user"
$ dnf install kernel-tools
$ cpupower frequency-set -g performance
```

List active services:

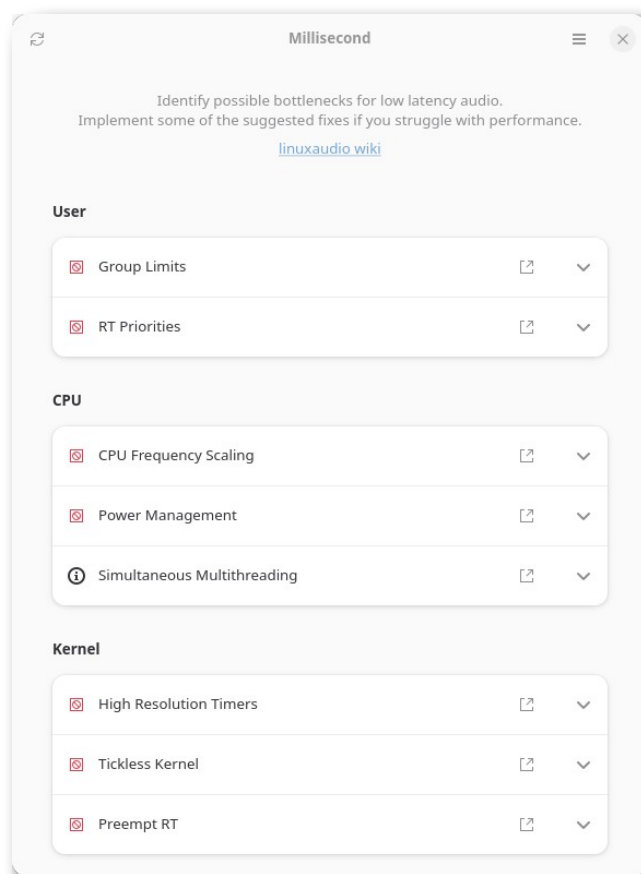
```
# Become a "super user"
$ su
# List all services
$ systemctl list-files
# List only active services
$ systemctl list-installed-files | grep -i enabled
# Disable certain services
$ systemctl {stop | disable} <service>
```



Fedora

Install the millisecond tool which allows you to check the MAO configuration of your Linux installation

```
$ dnf install millisecond  
$ millisecond
```





Fedora

Install RTirq: (via <http://linuxmao.org>)

RTirq is a bash script, intended to be started automatically when initializing the system (this type of program is called a "service").

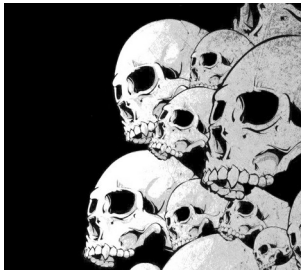
It is used in conjunction with a time-real core to increase the time-real-level priority of the IRQ threads, in order to optimize the latency of the peripherals (Audio interfaces).

Increasing the priority of IRQ threads is a compulsory step when using a time-real core with Jack.

You must add the optional threadirqs option when starting the kernel.

We install and start Rtirq before each audio session:

```
$ su
# Install RTirq
$ dnf install rtirq
# Start Rtirq
$ systemctl {start | enable} rtirq.service
# 1st choice: starting in the spot
$ systemctl start rtirq.service
# 2nd choice: activation of the RTirq service
$ systemctl enable rtirq.service
```



Fedora

Disable Packagekit

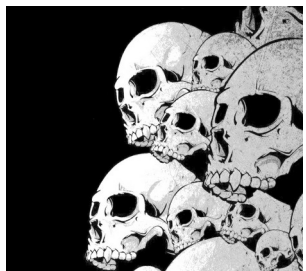
Packagekit is a system that allows you to check the presence of updates.

The concern: when it turns, it uses a lot of the hard drive (reconstruction of the database).
Another concern: the system system Disable Packagekit.service does not work after the reboot.

Tip: We will hide it ...

```
$ su  
$ systemctl mask packagekit.service
```

Connect the sound card to a USB 2.0 port



Fedora For Jack users

Modify the priorities of the Jack group:

```
$ su
$ cat /etc/security/limits.d/95-jack.conf
# Default limits for users of jack-audio-connection-kit

@jackuser - rtprio 90
@jackuser - nice -10
@jackuser - memlock unlimited
#@jackuser - memlock 4194304

#@pulse-rt - rtprio 10
#@pulse-rt - nice -20
```

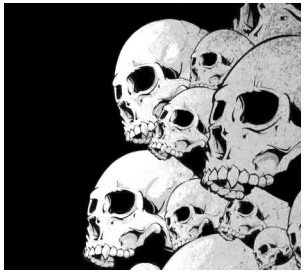
If you use
VCVRACK, leave
memory for video

These two lines
must be deleted
imperatively

Add yourself to the Jackuser group via:

```
$ usermod -a -G jackuser <username>
```

Then, disconnect and reconnect from your session.



Fedora For Pipewire / Jack users

Modify the priorities of the Jack group:

```
$ su
$ cat /etc/security/limits.d/25-pw-rlimits.conf

@pipewire - rtprio 70
@pipewire - nice -19
@pipewire - memlock unlimited
#@pipewire - memlock 4194304
```

If you use
VCVRACK, leave
memory for video

Add yourself to the Pipewire group via:

```
$ usermod -a -G pipewire <username>
```

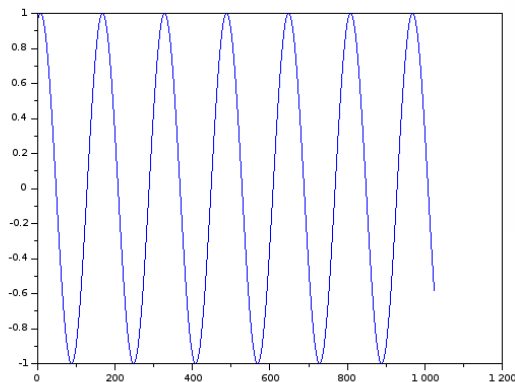
Then, disconnect and reconnect from your session

```
$ pw-metadata -n settings 0 clock.force-quantum 256
$ pw-metadata -n settings 0 clock.force-rate 44100
$ systemctl --user restart pipewire.service
$ pw-top
```

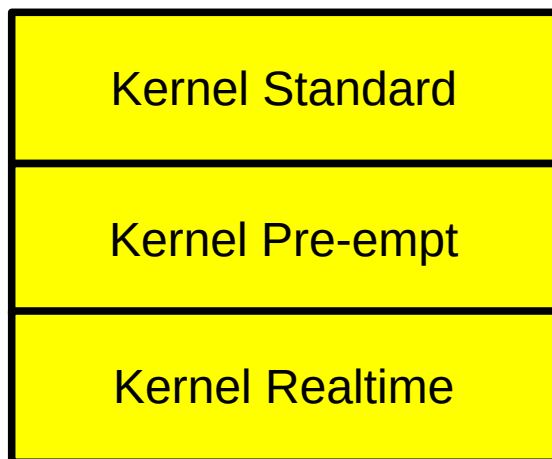
Some useful
command
lines



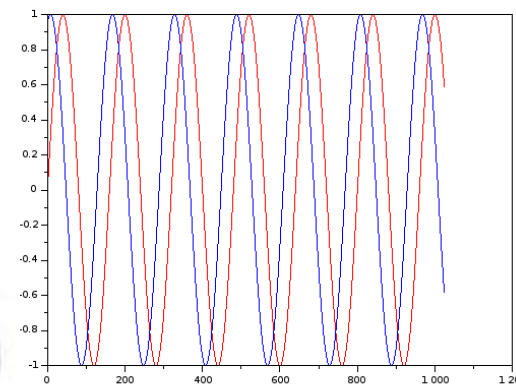
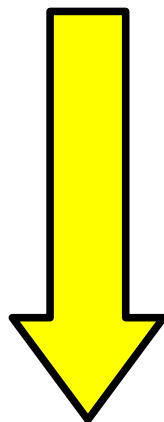
Kernel latency



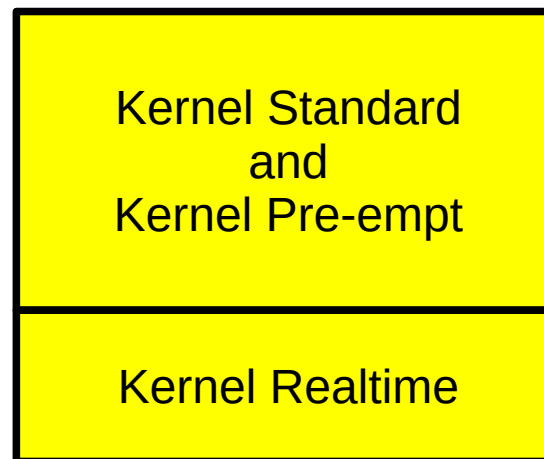
Avant



Decreasing latency



Today



The latency is tolerable to the ear below 10 ms



Kernels Alternatives

Kernel liquorix (Intel Only) - <https://liquorix.net>:

```
$ dnf install kernel-lqx-mao
```

Kernel Xanmod- <https://xanmod.org>:

```
$ dnf install kernel-xan-mao
```

Kernel Real Time - <https://wiki.linuxfoundation.org/realtime/start>:

```
$ dnf install kernel-rt-mao # Based on the last Kernel
```

```
$ dnf install kernel-rt-stable-mao # Based on Kernel 5.10
```

Boot time options (select the kernel to boot and press 'e' and add the following options):

```
preempt=full threadirqs
```

Install preempt kernel options via the kernel-audio-tuned package

```
$ dnf install kernel-audio-tuned
```